CLASS 9 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): Isotopes of an element have similar chemical properties. Reason (R): Isotopes have the same number of electrons. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A metre scale has a worn end and reads 0.3 cm at the start. A measured length reads 5.7 cm. The corrected length is: 1

(a) 5.4 cm **(b)** 5.7 cm **(c)** 6.0 cm **(d)** 0.3 cm

3. The solubility of KNO3 is 64 g/100 g water at 40 degrees Celsius and 32 g at 20 degrees Celsius. 164 g of saturated solution at 40 degrees Celsius is cooled to 20 degrees Celsius. The mass of crystals is: 1

(a) 96 g **(b)** 32 g **(c)** 16 g **(d)** 64 g

4. A sample of a compound contains 2.4 g of carbon and 6.4 g of oxygen. The ratio C : O by mass and the likely formula (C = 12, O = 16) are: 1

(a) 3 : 16, CO_3 **(b)** 3 : 8, CO_2 **(c)** 1 : 2, CO_2 **(d)** 3 : 4, CO

5. Two ions X^{2+} and Y^{2-} each have 18 electrons. The atomic numbers of X and Y are respectively: 1

(a) 19 and 17 **(b)** 20 and 16 **(c)** 18 and 18 **(d)** 16 and 20

6. Which situation requires an exact value rather than an estimate? 1

(a) dosage of a medicine for a child **(b)** number of stars in a galaxy
(c) height of a mountain in rough terms **(d)** population of a country

7. Which of the following combinations can be separated by magnetic separation? 1

(a) iron and sulphur **(b)** salt and sugar **(c)** sand and chalk **(d)** water and oil

8. An element X forms a chloride XCl_2 with 52.5 per cent X by mass? (Cl = 35.5) The atomic mass of X is closest to: 1

(a) 24 **(b)** 78.5 **(c)** 40 **(d)** 56

9. A sample of boron (average mass 10.8 u) has isotopes 10 and 11. The percentage of the lighter isotope is: 1

(a) 20 **(b)** 50 **(c)** 10.8 **(d)** 80

10. Assertion (A): Estimation is useful in science. Reason (R): It gives a quick check on whether an answer is reasonable. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): Fractional distillation is used to separate the components of air. Reason (R): The gases in liquid air have different boiling points. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. Assertion (A): Sodium chloride conducts electricity in the molten state but not in the solid state. Reason (R): In the molten state, the ions are free to move. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. **Assertion (A):** In Rutherford's experiment most alpha particles passed straight through the gold foil. 1
Reason (R): Most of the space inside an atom is empty.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** The least count of a metre scale marked in millimetres is 1 mm. 1
Reason (R): The metre is the SI unit of length.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** Fractional distillation is used to separate two miscible liquids whose boiling points differ by less than 25 K. 1
Reason (R): Simple distillation can separate a volatile liquid from a non-volatile dissolved solid.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Explain why the thermometer reading stays constant during distillation of pure water. 2

17. Write the electronic configurations of Na^{+} and F^{-} and say what they have in common. 2

18. State the reason why the first shell is filled with only two electrons. 2

19. Convert 1 light year into kilometres, given speed 3 x 10^5 km/s and 1 year = 3.15 x 10^7 s. 2

20. Why is the sugar crystal formed by slow cooling larger than that formed by rapid cooling? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. An atom has 5 more neutrons than protons and its mass number is 31. Find Z, n and the electronic configuration. 3

22. Explain how a graph of time period against length for a pendulum is expected to look. 3

23. Explain the 'salting out' of a colloidal sol such as soap using sodium chloride. 3

24. Find the percentage of carbon and of hydrogen in methane, CH4 (C = 12, H = 1). 3

OR

Explain how the Bohr model accounts for the observation that electrons do not fall into the nucleus. 3

25. The time for 30 oscillations of a pendulum is measured three times: 54.2 s, 54.6 s and 54.4 s. Find the mean time period and the percentage spread. 3

OR

In a chromatogram, three components X, Y, Z travel 2.0 cm, 4.5 cm and 6.0 cm while the solvent travels 7.5 cm. Find their R_f values and name the most strongly adsorbed. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** A student in Pune dissolves 20 g of common salt in 180 g of water to prepare a solution for an experiment. 5

(i) Name the solute and the solvent.

(ii) Find the mass of the solution.

(iii) Find the mass percentage of salt in the solution.

(iv) How much more water must be added to this solution to make a 5 per cent (by mass) solution?

27. Explain why equal masses of different elements do not contain equal numbers of atoms. Compare 12 g of carbon with 12 g of magnesium (C = 12, Mg = 24) in terms of the numbers of atoms. 5

OR

The nuclei of two elements A and B have the same number of neutrons (isotones), A has 12 protons and 24 u mass. Find the number of neutrons, the mass number of B if it has 11 protons, and the electronic configuration of B. 5

END OF QUESTION PAPER — Total: 50 Marks